

$$i) \quad r=3 \quad y=-4 \quad z=-7$$

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \\ &= \sqrt{(3)^2 + (-4)^2} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \phi &= \tan^{-1}(y/x) \\ &= \tan^{-1}(-4/3) \\ &= -53.130 \end{aligned}$$

$$\boxed{z = -7}$$

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} = \sqrt{(3)^2 + (-4)^2 + (-7)^2} \\ &= 8.352 \end{aligned}$$

$$\begin{aligned} \theta &= \cos^{-1}(z/r) = \cos^{-1}\left[\frac{-7}{8.352}\right] \\ &= 100.321 \end{aligned}$$

$$\begin{aligned} \phi &= \tan^{-1}(y/x) = \tan^{-1}\left[\frac{-4}{3}\right] \\ &= -53.130 \end{aligned}$$

$$ii) \quad \rho=3 \quad \phi=120^\circ \quad z=-5$$

$$\begin{aligned} x &= \rho \cos \phi = 3 \cos(120^\circ) \\ &= -1.5 \end{aligned}$$

$$\begin{aligned} y &= \rho \sin \phi = 3 \sin(120^\circ) = 2.598 \\ z &= -5 \end{aligned}$$

$$\begin{aligned} r &= \sqrt{\rho^2 + z^2} = \sqrt{(3)^2 + (-5)^2} \\ &= 5.831 \end{aligned}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{-5}{5.831}\right) = 100.321^\circ$$

$$\phi = \phi = 120^\circ$$

(iii)

$$r = 10 \quad \theta = -4 \quad \phi = -7$$

$$\begin{aligned} x &= r \sin \theta \cos \phi = 10 \cos \phi - 4 \\ &= 10 \cos -4 \sin -7 \\ &= -1.215 \end{aligned}$$

$$\begin{aligned} y &= r \sin \theta \sin \phi = 10 \sin -4 \sin -7 \\ &= 0.0850 \end{aligned}$$

$$z = r \cos \theta = 10 \cos -4 = 9.97$$

$$\begin{aligned} \rho &= r \sin \theta = 10 \sin -4 = -0.69 \\ \phi &= \phi = -7 \end{aligned}$$

~~$$\begin{aligned} z &= r \cos \theta \\ &= 10 \cos -4 \\ &= 9.97 \end{aligned}$$~~